SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology . (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I V Thanisha / 1965260160 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of student: V Thanisha

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

Annexure A - Scenario-Based Questions and Answers

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned: School of Computing /25; School of Electronics /26; School of Mechanical Engineering /27. As the network engineer, determine each network and broadcast address, valid host range, usable hosts, and remaining unused IP range.

Answer

Step 1: Given 192.168.100.0/24, total addresses = 2^(32-24) = 256. Allocate largest subnet first to prevent overlap.

Step 2: School of Computing: /25 = 255.255.255.128; block size 128. Network 192.168.100.0; broadcast 192.168.100.127; valid hosts 192.168.100.1 to 192.168.100.126; usable hosts = 2^7 - 2 = 126.

Step 3: School of Electronics: /26 = 255.255.255.192; block size 64. Network 192.168.100.128; broadcast 192.168.100.191; valid hosts 192.168.100.129 to 192.168.100.190; usable hosts = 2^6 - 2 = 62.

Step 4: School of Mechanical Engineering: /27 = 255.255.255.224; block size 32. Network 192.168.100.192; broadcast 192.168.100.223; valid hosts 192.168.100.193 to 192.168.100.222; usable hosts = 2^5 - 2 = 30.

Step 5: Allocated = 128 + 64 + 32 = 224 addresses. Remaining = 256 - 224 = 32 addresses: 192.168.100.224 to 192.168.100.255 (192.168.100.224/27). If used as a /27, 30 host addresses are usable.

Result: All three department subnets are distinct and fit within the original /24 block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Subnet masks: Software Development /24, Quality Assurance /25, Human Resources /26, Executive Management /27. Determine each department network address, broadcast address, valid host range, usable hosts, and remaining unused address range.

Answer

Step 1: The /16 block contains 2^(32-16) = 65,536 addresses, from 10.10.0.0 through 10.10.255.255. Allocate in descending order of subnet size.

Step 2: Software Development /24: network 10.10.0.0; broadcast 10.10.0.255; hosts 10.10.0.1 to 10.10.0.254; usable = 2^8 - 2 = 254.

Step 3: Quality Assurance /25: network 10.10.1.0; broadcast 10.10.1.127; hosts 10.10.1.1 to 10.10.1.126; usable = 2^7 - 2 = 126.

Step 4: Human Resources /26: network 10.10.1.128; broadcast 10.10.1.191; hosts 10.10.1.129 to 10.10.1.190; usable = 2^6 - 2 = 62.

Step 5: Executive Management /27: network 10.10.1.192; broadcast 10.10.1.223; hosts 10.10.1.193 to 10.10.1.222; usable = 2^5 - 2 = 30.

Step 6: Allocated addresses = 256 + 128 + 64 + 32 = 480. Unallocated = 65,536 - 480 = 65,056 addresses. Remaining contiguous range: 10.10.1.224 to 10.10.255.255.

Result: This VLSM allocation uses non-overlapping subnets and leaves 65,056 addresses for expansion.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The assigned network is 2001:0db8:abcd:0000:0000:0000:0000:0000/64. Convert it to compressed notation, expand it, identify the /64 prefix, and calculate the total possible host addresses.

Answer

Step 1: Given IPv6 address 2001:0db8:abcd:0000:0000:0000:0000:0000/64. IPv6 addresses contain 128 bits in eight groups of four hexadecimal digits.

Step 2: Remove leading zeroes in each group: 2001:db8:abcd:0:0:0:0:0.

Step 3: Replace the consecutive zero groups once with ::. Compressed address = 2001:db8:abcd::/64.

Step 4: Expand by restoring five all-zero groups after abcd: 2001:0db8:abcd:0000:0000:0000:0000:0000/64.

Step 5: A /64 prefix reserves the first 64 bits (four groups) for the network. Network prefix = 2001:0db8:abcd:0000::/64, or 2001:db8:abcd:0::/64.

Step 6: Host/interface-identifier bits = 128 - 64 = 64. Total possible interface identifiers = 2^64 = 18,446,744,073,709,551,616. Unlike IPv4, do not automatically subtract two for network and broadcast addresses; IPv6 has no broadcast address.

Result: The /64 subnet has 2^64 possible interface identifiers. Some values may be reserved or have special uses, so this is the theoretical address-space count.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. Routing entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24. A packet arrives for 192.168.2.45. Determine the destination subnet, matching route, next-hop network/interface, and the default route if no entry matches.

Answer

Step 1: Given destination = 192.168.2.45. Each /24 mask is 255.255.255.0, so the first three octets identify the network.

Step 2: Perform bitwise AND: 192.168.2.45 AND 255.255.255.0 = 192.168.2.0. Destination subnet = 192.168.2.0/24.

Step 3: Compare the routing table entries. 192.168.1.0/24 does not match; 192.168.2.0/24 matches; 192.168.3.0/24 does not match. The matching entry is 192.168.2.0/24.

Step 4: The router forwards the packet using the outgoing interface or next-hop associated with 192.168.2.0/24. The question does not provide interface names or next-hop IPs, so a specific next-hop address cannot be determined.

Step 5: If no specific entry matches, a configured default route 0.0.0.0/0 would be selected. Its actual next-hop must be supplied by the ISP/router configuration; none is given.

Step 6: Verify forwarding with show ip route, show ip interface brief, ping and traceroute in a simulation; confirm the route is installed and the interface is operational.

Result: Route 192.168.2.0/24 is the correct match. The exact forwarding interface and default-route next hop are unspecified.

5. A company establishes two office departments connected through a Layer-3 router. Administration LAN = 192.168.10.0/26, Finance LAN = 192.168.10.64/26. Router gateways: Administration = 192.168.10.1 and Finance = 192.168.10.65. Determine network and broadcast addresses, valid host ranges, suitable addresses for three computers in each LAN, and their default gateways.

Answer

Step 1: Both LANs use /26 = 255.255.255.192. Block size = 256 - 192 = 64 addresses. Each /26 has 2^6 - 2 = 62 usable IPv4 host addresses.

Step 2: Administration: network 192.168.10.0; broadcast 192.168.10.63; host range 192.168.10.1 to 192.168.10.62; gateway 192.168.10.1.

Step 3: Finance: network 192.168.10.64; broadcast 192.168.10.127; host range 192.168.10.65 to 192.168.10.126; gateway 192.168.10.65.

Step 4: Assign Administration PCs 192.168.10.2, 192.168.10.3 and 192.168.10.4, each with mask 255.255.255.192 and default gateway 192.168.10.1.

Step 5: Assign Finance PCs 192.168.10.66, 192.168.10.67 and 192.168.10.68, each with mask 255.255.255.192 and default gateway 192.168.10.65.

Step 6: Check that PC IPs are unique, not network/broadcast addresses, and do not duplicate the gateways. Router interfaces must be active and routing between connected LANs enabled. Test connectivity with ping.

Result: Both LANs have 62 usable addresses, valid gateways, and three non-conflicting example computer addresses each.